

Concrete examples of revision behavior

Date prepared: 2026-07-13

Run: 20260710T161217Z

Reported model: llama-3.1-8b-instant

Purpose and reading guide

These examples ground the aggregate analysis in individual PubMedQA traces. In each case, Stage 1 answered from the evidence available before the first RESULTS section (or before the fallback half split). Stage 2 then saw the added evidence and returned one of KEEP_ANSWER, REVISE_ANSWER, or ABSTAIN.

The question text comes from [data/pubmedqa/ori_pqal.json](#); the trace files record the evidence and outputs. They do **not** record a model rationale. The explanations below are therefore analyst interpretations of the visible evidence, not claims about the model's hidden reasoning.

Example index

Behavior	PMID	Gold	Stage 1	Stage 2 action	Final	Why it matters
Successful correction	11138995	no	yes	revise	no	Added results directly reject the broad risk-factor claim.
Successful correction from uncertainty	15919266	no	maybe	revise	no	Comparative outcomes show no benefit from adjuvant radiation.
Correct retention	10158597	yes	yes	keep	yes	The main benefit survives despite important secondary caveats.
Missed correction	10201555	yes	maybe	keep	maybe	Added adjusted effect estimates support a yes answer.
Harmful overreaction	10548670	yes	yes	revise	no	Added quantitative evidence supports, rather than overturns, Stage 1.
Wrong-direction revision	19302863	yes	maybe	revise	no	Stage 2 changes the answer but still moves away from the gold label.
Ambiguity collapsed to no	11867487	maybe	maybe	revise	no	Added evidence concerns beliefs and use, not efficacy.
Abstention despite directional evidence	17054994	no	yes	abstain	--	Added results clearly say the procedure never changed management.

Successful revisions

PMID 11138995: a direct negative result corrects an affirmative prior answer

Question: Is alexithymia a risk factor for unexplained physical symptoms in general medical outpatients?

Observed path: yes -> REVISE_ANSWER -> no (gold: no).

What changed: Stage 1 saw the study framing and methods and answered yes. The added results reported that patients with medically unexplained symptoms were not more alexithymic overall, and that alexithymia was not associated with subjective health or service use. A narrower subgroup association remained, but it did not support the broad question.

Interpretation: This is the intended revision behavior. Stage 2 uses outcome evidence to replace a hypothesis-consistent answer with the study's broader null result. It also shows why the question's scope matters: a subgroup signal should not override the overall finding.

Source: [trace_11138995.json](#)

PMID 15919266: uncertainty resolves to no after a comparative null result

Question: Adjuvant radiation of stage III thymoma: is it necessary?

Observed path: maybe -> REVISE_ANSWER -> no (gold: no).

What changed: The added evidence reported no alteration in local or distant recurrence and similar 10-year disease-specific survival with and without radiation (79% versus 75%, p = 0.21).

Interpretation: Stage 2 appropriately converts uncertainty into a negative answer when the comparative results do not demonstrate benefit. This is a stronger example of evidence-based revision than merely switching because a RESULTS section contains negative language.

Source: [trace_15919266.json](#)

Appropriate maintenance

PMID 10158597: keeping the main answer while tolerating caveats

Question: Does a dedicated discharge coordinator improve the quality of hospital discharge?

Observed path: yes -> KEEP_ANSWER -> yes (gold: yes).

What changed: The added results showed improved discharge planning and fewer post-discharge problems, while finding no improvement in the timing of community services or bed-use efficiency.

Interpretation: Keeping yes is reasonable because the primary claim is supported even though not every secondary outcome improved. This trace demonstrates that a good revision policy must integrate mixed evidence rather than treating any caveat as a reason to reverse.

Source: [trace_10158597.json](#)

Missed and harmful revisions

PMID 10201555: strong new evidence is ignored

Question: Is low serum chloride level a risk factor for cardiovascular mortality?

Observed path: maybe -> KEEP_ANSWER -> maybe (gold: yes).

What changed: The added results described serum chloride as an independent predictor after adjustment, with elevated cardiovascular-death risk for low chloride in both men and women and a dose-response relationship.

Interpretation: This is a clear missed opportunity. The added evidence supplies the adjusted association and effect estimates needed to move from maybe to yes, yet Stage 2 leaves the preliminary answer unchanged.

Source: [trace_10201555.json](#)

PMID 10548670: added evidence triggers the opposite of the supported answer

Question: Does the National Institutes of Health Stroke Scale favor left hemisphere strokes?

Observed path: yes -> REVISE_ANSWER -> no (gold: yes).

What changed: At comparable NIH Stroke Scale scores below 20, right-hemisphere infarcts were approximately twice as large as left-hemisphere infarcts. That pattern supports the proposition that the scale gives relatively more weight to deficits associated with left-hemisphere strokes.

Interpretation: The revision is harmful and difficult to justify from the visible result. A plausible failure mode is losing the direction of a comparative relationship: "right lesions are larger at the same score" is evidence of left-hemisphere score bias, not evidence against it.

Source: [trace_10548670.json](#)

PMID 19302863: revising is not the same as correcting

Question: Is the use of cyanoacrylate in intestinal anastomosis a good and reliable alternative?

Observed path: maybe -> REVISE_ANSWER -> no (gold: yes).

What changed: The added evidence found no significant difference in bursting pressure and reported better hydroxyproline levels and anastomosis time for the cyanoacrylate subgroups.

Interpretation: Stage 2 reacts to the new evidence but chooses the wrong direction. This trace is counted as a missed revision, not an overreaction, because Stage 1 was already wrong. It illustrates why action counts alone are inadequate: REVISE_ANSWER can leave accuracy unchanged or make the semantic interpretation worse.

Source: [trace_19302863.json](#)

Ambiguity and abstention

PMID 11867487: non-efficacy evidence is treated as evidence of no efficacy

Question: Does rugby headgear prevent concussion?

Observed path: maybe -> REVISE_ANSWER -> no (gold: maybe).

What changed: The added evidence measured player and coach beliefs, headgear use, and reasons for non-use. It did not test whether headgear actually prevents concussion.

Interpretation: maybe remains the defensible answer because the added section does not resolve the causal question. The revision appears to conflate weak or indirect evidence with negative evidence.

Source: [trace_11867487.json](#)

PMID 17054994: abstention is used where the result is unusually clear

Question: Does frozen section alter surgical management of multinodular thyroid disease?

Observed path: yes -> ABSTAIN -> no final answer (gold: no).

What changed: The added results reported 25% sensitivity and, most directly, that frozen-section analysis altered intraoperative management in none of 135 patients.

Interpretation: Abstention avoids retaining the wrong yes, but it also fails to convert highly directional evidence into the correct no. Because all three abstentions in the run had gold label no, the small abstention set is more consistent with unresolved negative evidence than with a calibrated, label-neutral reject option.

Source: [trace_17054994.json](#)

Cross-example pattern

Taken together, the examples support three conclusions from the aggregate results:

1. Stage 2 can correctly exploit explicit null or negative findings, especially when revising yes or maybe to no.
2. It sometimes treats caveats, indirect evidence, or added complexity as a generic signal to move toward no, even when the evidence supports yes or remains inconclusive.
3. Many correctable Stage 1 errors remain untouched. The main limitation is not only harmful revision; it is failure to revise when added evidence is decisive.

These examples were selected to illustrate distinct behaviors, not to estimate their prevalence. Prevalence and class-level effects are reported in [latest-run-results.md](#).